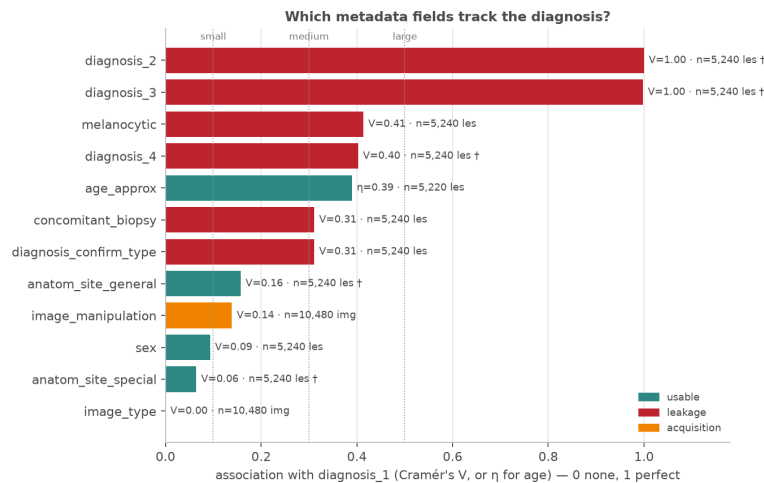


Session 2 homework — Extended EDA & data pipeline (MILK10k)

Stanislaus Lattorff · Computer Vision & Speech Recognition, EADA · September 2026 Code: <https://github.com/stansilaisjfnep/milk10k-computer-vision> — src/milk10k/eda2.py (Parts 1–2), pipeline.py (3–4), viz.py (5); python scripts/run_session2.py regenerates every number and figure below.

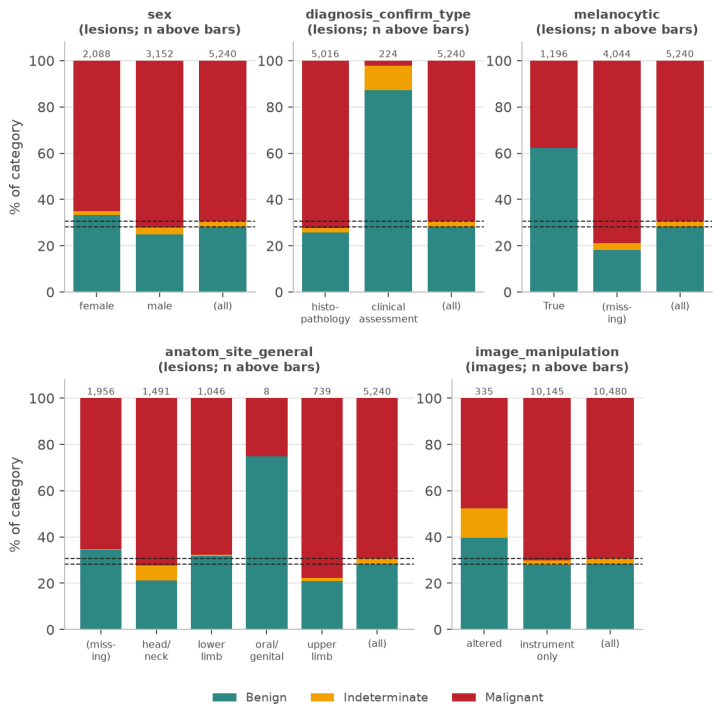
Part 1 — Which metadata fields relate to the diagnosis?

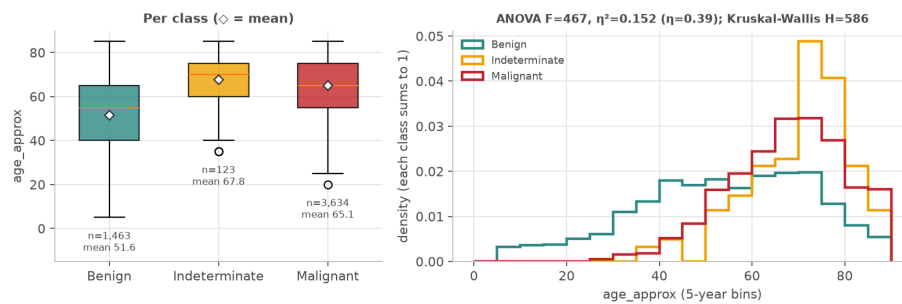
- Unit of analysis = the lesion.** metadata.csv has 10,480 rows but 5,240 lesions, each photographed twice. All fields except image_type and image_manipulation are identical on a lesion's two images, so testing image rows would count every lesion twice and double every χ^2 . Lesion-constant fields are tested on 5,240 lesions, the two image-level fields on 10,480 images.
- Statistics.** Categorical vs. the 3-class target: χ^2 test of independence + Cramér's V (V rescales χ^2 to 0–1 whatever the number of categories, so fields are comparable). Numeric age_approx : one-way ANOVA + η (share of variance explained by class; η is on the same 0–1 scale as V), with Kruskal–Wallis as a rank-based check because age is binned and skewed. At n = 5,240 every p-value is ≈ 0 , so fields are ranked by effect size, not by significance. Missing values are kept as their own category, because whether a field was recorded turns out to be informative.



† chi-square p-value unreliable (sparse cells); V still describes the table. les = tested on 5,240 lesions, img = on 10,480 images.

Field	Role	Effect	Reading
diagnosis_2, diagnosis_3	leakage	V = 1.00	finer levels of the diagnosis — they are the label
melanocytic	leakage	V = 0.41	only ever filled in for nevi and melanomas; its presence names the lesion family
age_approx	usable	$\eta = 0.39$ ($\eta^2 = 0.15$)	benign lesions: mean 52 y; malignant 65 y; indeterminate 68 y
diagnosis_confirm_type = concomitant_biopsy	leakage	V = 0.31	identical variables; non-biopsied lesions are 2 % malignant (5 of 224)
anatom_site_general	usable	V = 0.16	head/neck carries most Indeterminate lesions; 37 % missing
image_manipulation	acquisition	V = 0.14	"altered" photos are 13 % Indeterminate vs. 2 % overall
sex, anatom_site_special	usable	V = 0.09 / 0.06	negligible
image_type	acquisition	V = 0.00	every lesion has one of each, so independent by construction
attribution, copyright_license	constant	—	a single value for all rows



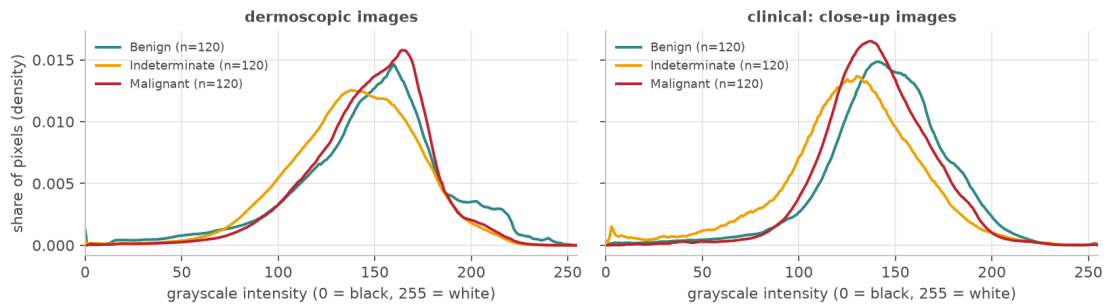


- **Most associated usable fields:** age (medium effect), then anatomic site (small). **Least useful:** sex and anatom_site_special (negligible), image_type (zero by design) and the two constant columns.
- **Leakage — never a model input:** diagnosis_2/3/4 contain the answer; diagnosis_confirm_type/concomitant_biopsy record *that a biopsy was done*, a decision made because a clinician was already worried, and the rare lesions that were not biopsied are almost all benign; melanocytic leaks through its missingness.
- **Bias / shortcut risk:** image_manipulation describes how the photograph was produced, not the lesion, yet shifts the class mix; a model could learn "edited photo → indeterminate". Age and site are legitimate but also mark the *population* (older, sun-exposed patients), so a model must not reduce to "old + face → malignant".

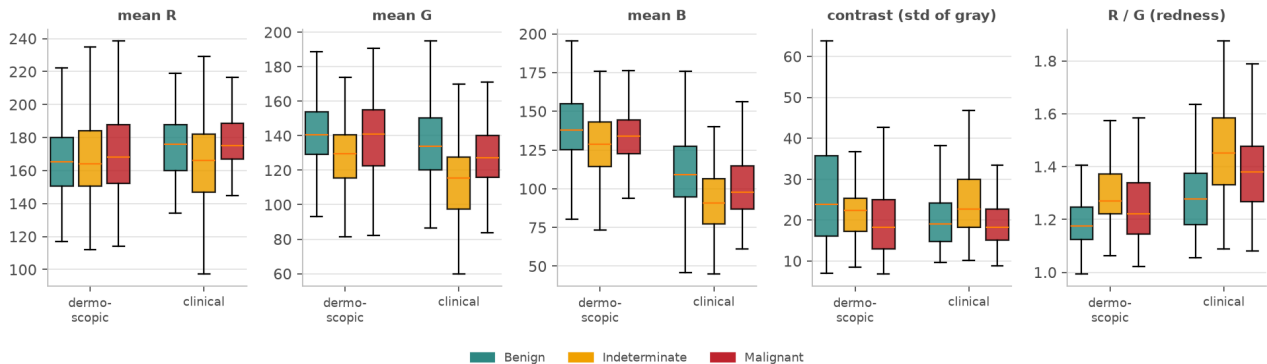
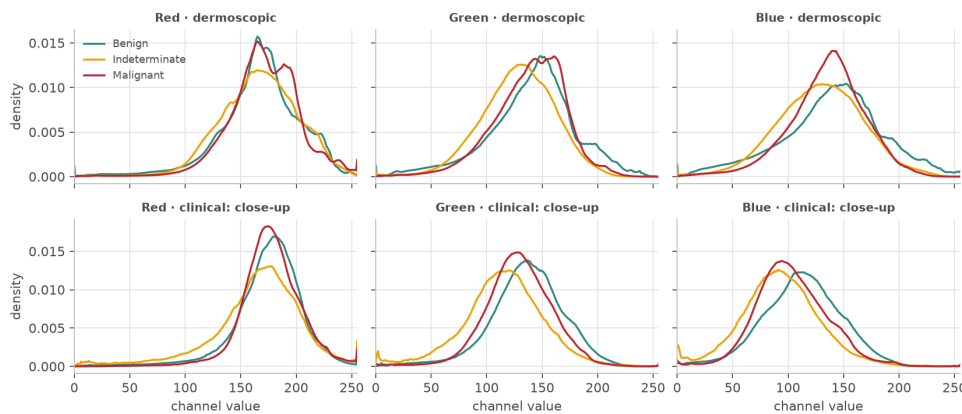
Part 2 — Colour and histograms across classes

- **Sample.** 120 lesions per class (Indeterminate has only 123), **both images of each** → 720 images at full 600×450 resolution, balanced so class imbalance cannot drive any difference. Histograms are normalised per image before averaging, so every image weighs the same.
- **Split by modality.** Dermoscopy (contact lens, own light source) and clinical close-ups are shown separately; pooling them blurs two different pictures: for all nine colour features the class effect (η^2) is smaller pooled than within clinical images.

Average grayscale histogram per class



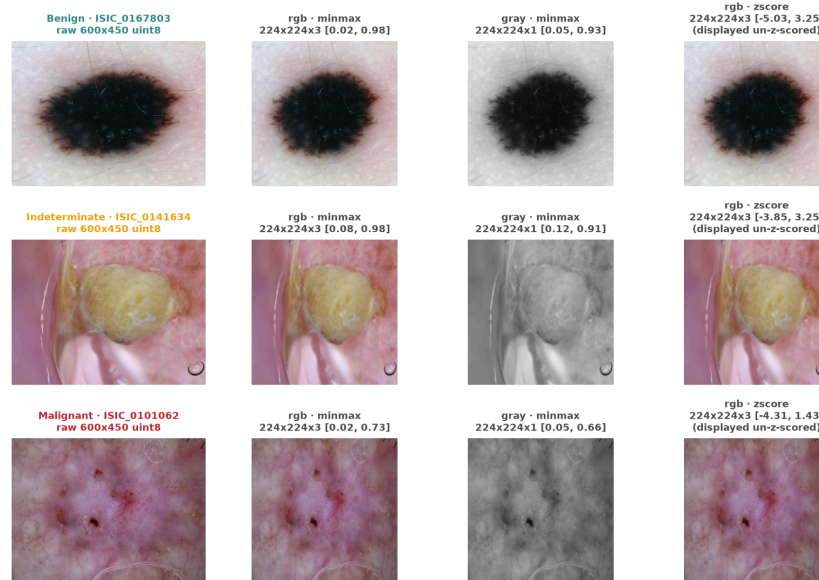
Average per-channel histogram per class



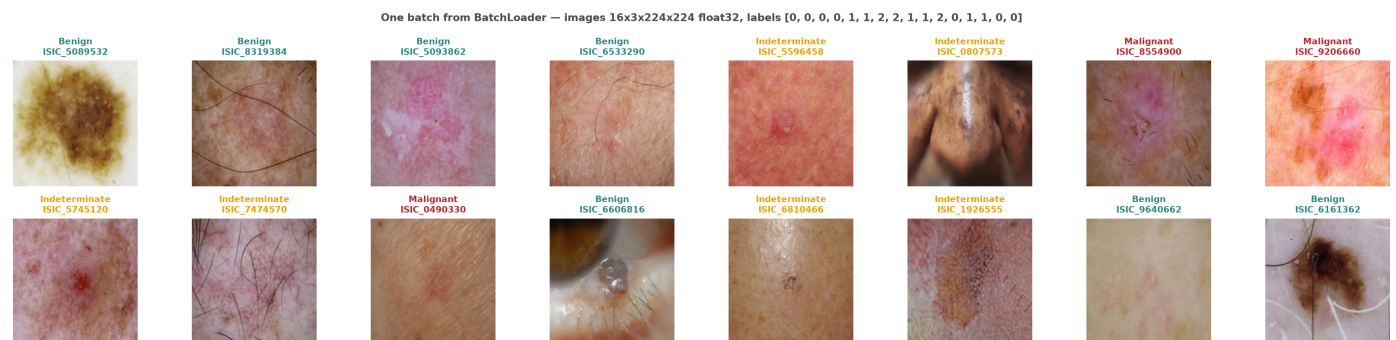
Per image (mean ± sd)	Dermoscopic: Benign	Indet.	Malignant	Clinical: Benign	Indet.	Malignant
mean gray (brightness)	148 ± 22	140 ± 22	148 ± 20	145 ± 21	127 ± 22	140 ± 19
mean G	141 ± 22	128 ± 20	139 ± 20	136 ± 23	114 ± 23	129 ± 20
R / G (redness)	1.2 ± 0.1	1.3 ± 0.1	1.2 ± 0.1	1.3 ± 0.2	1.5 ± 0.2	1.4 ± 0.2
std gray (contrast)	28 ± 15	23 ± 9	21 ± 11	21 ± 10	26 ± 12	19 ± 6

- **What differs:** Indeterminate images are darker and redder, clearly so in clinical photos. Benign dermoscopic images show a tail of bright pixels (190–230) and higher contrast. Benign and Malignant are close to each other in every channel.
- **Alternative explanations — the difference is not necessarily the lesion.** In this very sample, Indeterminate lesions are **80 % head/neck** (Benign 20 %, Malignant 28 %), **15 % Fitzpatrick type I** skin (others ≤ 1 %) and **17.5 % "altered"** photos (others 2–5 %): sun-damaged, fair facial skin photographed under different conditions. Five of the 120 Indeterminate clinical photos contain large dark backgrounds, which is where the spike at 0 in the clinical grayscale histogram comes from. Body site, skin type and photo production all change colour before any diagnosis does. The clinics and cameras cannot be checked directly — attribution is identical for every image — which is itself a limitation.
- **Is colour alone a useful signal? Weak, and partly confounded.** No colour feature explains more than 15 % of its variance by class ($\eta^2 \leq 0.15$; ≤ 0.12 in dermoscopy). A logistic regression on all nine colour statistics reaches a **balanced accuracy of 0.47 on dermoscopic and 0.56 on clinical images** (5-fold CV, chance 0.33) — better than chance, far from useful, and strongest exactly where the site/skin-type confounding is strongest. The classes overlap heavily; the project needs a model that learns **spatial structure** (borders, texture, pigment networks), i.e. a CNN, with metadata such as age and site added carefully rather than colour thresholds.

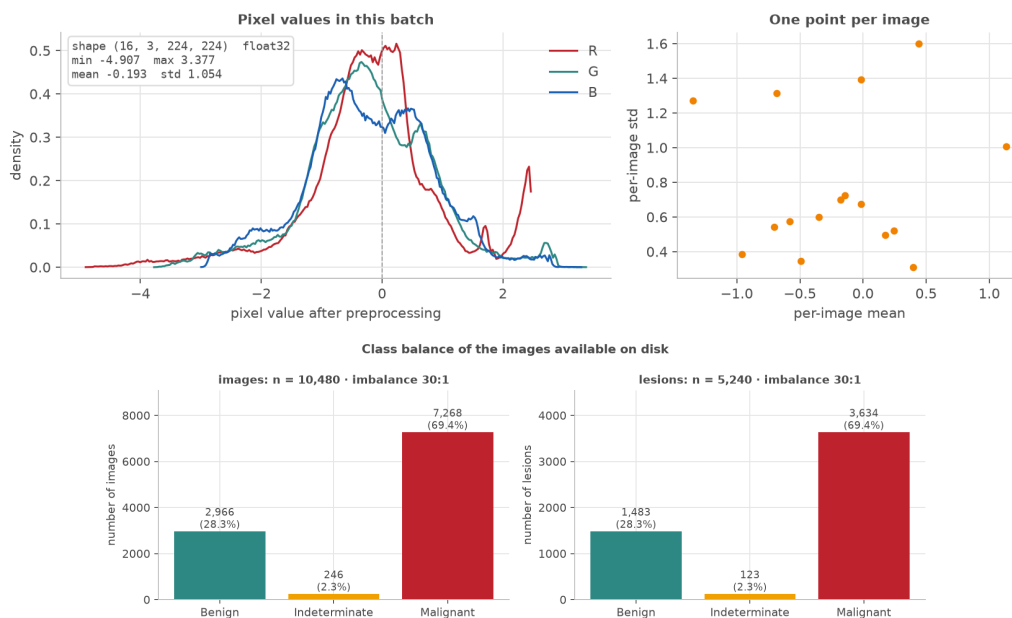
Parts 3–5 — Reusable pipeline



- `process_image(source, size=224, color_space="rgb"|"gray"|"hsv"|"lab", normalize="minmax"|"zscore"|"none", channels_first=False)` accepts a path, an `isic_id`, a PIL image or an array and returns a `ProcessedImage`: the array plus `shape`, `dtype`, `value_range`, and `steps` (e.g. `resize 600x450 → 224x224 → rgb → minmax → HWC → CHW`). **Default normalisation is min-max with the encoding's fixed bounds (x / 255)**: it needs no fitted statistics, so it cannot leak across splits, and unlike per-image min-max it keeps the brightness differences Part 2 measured. Z-score uses the **training-split** mean/std from `artifacts/norm_stats.json` by default (or ImageNet's for a pretrained backbone).
- `process_batch(paths | ids | DataFrame, **same options)` never stops on one bad file: a truncated JPEG, a non-image and a missing id are each returned in `skipped` as `(id, "ExceptionType: message")` while the good images are stacked (tested in `tests/test_session2.py`).



```
BatchLoader(metadata, img_dir=None, *, batch_size=32, label_col="diagnosis_1",
             classes=("Benign", "Indeterminate", "Malignant"), shuffle=False, seed=42,
             drop_last=False, channels_first=True, **process_image options)
len(loader)    -> batches per epoch      loader.n_missing / .missing_ids -> rows with no file on disk
for batch in loader: # LoaderBatch, read and processed on the fly (memory = one batch)
    batch.images      # float32 (B, 3, 224, 224)      batch.labels      # int64 (B,) class indices
    batch.label_names # ["Malignant", ...]           batch.ids         # isic_id per image
    batch.skipped     # [(isic_id, reason), ...] for files that failed in this batch
```



- **Loader and visualiser in use.** The grid above is a real batch, drawn through `viz.to_displayable`, which undoes z-scoring and reorders CHW→HWC before display. The batch summary confirms z-scoring worked (mean -0.19 , std 1.05) and exposes red pixels piling up at 255 ($z \approx 2.46$): saturation worth knowing about before augmentation. **Class balance is severe (30:1 at image and lesion level)**, so later training needs lesion-grouped stratified splits, class weights or a weighted sampler, and evaluation by balanced accuracy / per-class recall rather than accuracy, which a model predicting "Malignant" for everything would already score at 69 %.
- **Design choices.** Each part is one layer that calls the one below instead of copying it. `process_image` reuses the project's existing `resize` (`preprocessing.preprocess_image`, pixel-identical — a test enforces it) so analysis and training can never preprocess differently; `process_batch` only adds error handling around it; `BatchLoader` only adds the availability filter (one directory listing, not a file check per row), labels, shuffling and batching around `process_batch`; and every function in `viz` accepts exactly what the loader yields. Every setting is a keyword argument with a documented default, and every result carries a record of what was done, so later milestones can call these functions rather than re-derive them. The torch `DataLoader` in `datasets.py` stays the training path (augmentation, workers); this numpy loader is for analysis and debugging.